

Time Series Analysis of Carbon Dioxide Levels in Niwot Ridge, Colorado from 2012 through 2021

University of Maryland, Baltimore County Statistics Department

Tobi Akinyemi, Jayani Bakshi, Alex Caramanico, Brooke Springfield, Joanne Wang

11 December 2022

Table of Contents

Executive Summary.....	3
Analysis.....	4
Introduction.....	4
Exploratory Data Analysis.....	4
Time Series Regression.....	7
Trend Modeling.....	7
Seasonality Modeling.....	8
Residual Modeling.....	8
Combining Components.....	10
Exponential Smoothing.....	13
ARIMA Models.....	14
Forecasting.....	16
Conclusion.....	18
References.....	20
Appendix.....	21

Executive Summary

The following report examined data collected by the National Oceanic and Atmospheric Administration on carbon dioxide levels in Niwot Ridge, Colorado from 2012 through 2021 with the objective of projecting future values of the gas. To accomplish this, several time series analysis methods were utilized, including: time series regression, exponential smoothing, and autoregressive integrated moving average (ARIMA) models. Within each of these categories, the most appropriate forecasting model was selected. After which, the three were compared against each other in order to determine the most accurate and efficient model overall. The primary results are explained below. It should be mentioned that the initial ten-year data set was split into training and test data, with 90% used for model building and 10% used for model testing in order to calculate forecast errors.

For time series regression, the data was decomposed into three components: trend, seasonality, and residuals. Using the Ljung-Box test, the residuals of each component were tested for autocorrelation to ensure there was not any leftover pattern within them. For the selected carbon dioxide data set, the optimal trend model was found to be a linear function with respect to time, with both the slope and intercept parameters significant. Seasonality was ultimately modeled using several dummy variables, one representing each month of the year. Once the residuals for these models were assessed and corrected, the three components were added back together to create the overall time series model. At this point, the model was tested using the test data, from which forecast errors were calculated. The model performed reasonably well, with relatively low errors but slightly larger than the exponential smoothing and ARIMA errors. Lastly, the model was used to forecast 2022, predicting carbon dioxide levels to continue increasing while maintaining its seasonality. That is, the model predicted low values in the early fall months and high values in the spring months.

For exponential smoothing, the Holt-Winters method was implemented using R. This selected the optimal values for parameters α , β , and γ , which control the level, slope, and seasonality respectively. The implementation found α to be very close to 1, β as 0, and γ as 1. This means that while the slope does not update as time changes, the level and seasonality of the data are based on recent time points. Then, using the forecasting library, the Holt-Winters result gave predicted values for 2021 and 2022. The values for 2021 were used to calculate errors, whereas 2022 was a projection into the future. The method performed extremely well, with considerably low errors committed. Additionally, its predictions were very similar to that of the time series regression method.

Lastly, ARIMA models were utilized. In short, an explicit model was developed using the `auto.arima` function in R, based upon the autocorrelation function and partial autocorrelation function of the residuals. It found the parameters to be: $p = 5$, $d = 0$, $q = 1$. In other words, the model predicts future values involving the previous p time points and q white noise. It also found there to be two seasonal autoregressive terms after one difference of the data set. Each time point and white noise term had its respective coefficient estimated. This model was then tested using the test data, once again achieving very low errors. Finally, the model was used to forecast 2022, finding similar results to both the regression results and exponential smoothing results, strengthening the conclusion that carbon dioxide will continue to increase over time, maintaining its seasonal patterns. While the results may vary from location to location, the results of this investigation can be extrapolated to say recent carbon dioxide patterns are expected to continue.

I - Introduction

One of the most crucial issues facing the planet today is the issue of greenhouse gasses. While society continues to expand and develop, the emission of such gasses grows at an ever-increasing rate. The primary source of such emissions is the production of energy that is used in hundreds of day-to-day activities around the world. Energy accounts for 72% of all man-made greenhouse gas emissions, and can be broken down as follows: electricity and heat 31%, transportation 15%, manufacturing and construction 12%, agriculture 11%, and forestry 6% [1]. According to the Center for Climate and Energy Solutions, carbon dioxide made up approximately 76% of all greenhouse gas emissions in 2015. This extreme value may come as no surprise due to the rampant burning of fossil fuels, the manufacturing of cement, and from the combustion of fuels in millions of non-electric cars around the world [2]. Excess carbon dioxide in the atmosphere has several adverse effects, many of which have already been observed. The most trivial of these is the effect by which the gasses get their name: the trapping of heat leading to the increase of global temperatures, akin to a greenhouse. Increased temperatures can lead directly to the melting of polar ice caps, thereby raising sea levels. This can in turn lead to the destruction or alteration of ecosystems for countless species. Rising temperatures are directly linked to drastic climate change, increasing the volatility and often the hostility of weather. Since the effects of carbon dioxide emissions affect every corner of the planet, it is certainly an important problem to understand and to address.

There may not be a concrete solution to this problem, at least not one that is relatively inexpensive or logistically feasible, but an important first step is to understand the trend that has taken place over past years. This report will seek to model the change in carbon dioxide levels in Niwot Ridge, Colorado in order to make predictions on future levels of the gas. It will use data from the National Oceanic and Atmospheric Administration [3]. Several time series analysis techniques will be utilized, including regression, exponential smoothing, and autoregressive integrated moving average (ARIMA) models. The best forecasting method created by these techniques will be selected through various considerations, such as minimizing error functions and maximizing model simplicity. In addition to this, there will be a strong emphasis on the leftover residuals from the application of the analysis techniques, a facet which will be explained in greater detail in a later section.

While the report will not offer solutions to the problem, it will likely provide evidence for its pertinence supported through statistical analyses. The report therein will give readers a stronger understanding of carbon dioxide trends, perhaps eliciting a stronger desire for improvement.

II - Time Series Analysis

i. Exploratory Data Analysis

As mentioned, this report will focus on data collected from Niwot Ridge, California from 1967 through 2021. Each sample was collected from the air using glass flasks, with measurements taking place once every month. The carbon dioxide was detected by a non-dispersive infrared (NDIR) analyzer, after which dry air mole fractions were calculated for the gas [4]. While this is a somewhat complicated process to describe in detail, it should be mentioned that a higher mole fraction, or carbon dioxide level, means there is a larger amount of carbon dioxide in the air. Though the data set begins in 1967, there are some missing values between 1974 and 1975, so any data collected before 1976 will not be included. This still leaves a reasonable sample size with which to work.

Before conducting any analyses on the data, it is important to gain an initial understanding of the data set and to observe potential patterns. This will give more credibility to the eventual models and provide significantly more room for their interpretations. In other words, preliminary observations of the data may be supported or be opposed by the results of the analyses. To do this, the plotting function in R will be used to display the data in various ways.

Figure 1 below shows the entire data set plotted as a function of time, with the vertical axis representing the amount of carbon dioxide measured at each time point. While the plot appears continuous, it should be remembered that the measurements were not being taken continuously, but rather on a monthly basis. That said, connecting each time point in this way facilitates pattern observations. It is apparent that carbon dioxide levels have been steadily increasing since the beginning of the measurements. It can even be seen that the pattern is relatively consistent, that is carbon dioxide levels appear to increase at a constant rate rather than an exponential or logarithmic or quadratic rate. There is also clearly an element of seasonality, in which carbon dioxide fluctuates depending on the time of year.

This can be more easily seen with Figure 2, which shows the data from 2012 through 2021. It appears that the level of carbon dioxide in the air consistently reaches its annual low values around August or September, and reaches its annual high values around March or April. Based on this information, it can be said that the data shows clear trend and seasonality, which greatly improves its suitability for time series analysis.

Random noise can also be observed using this figure. Such areas can be seen in places where the function is not as smooth or as consistent. That is, the valley between 2015 and 2016 is deeper than the valley immediately following it between 2016 and 2017. This is to say that the low values of 2015 are lower than the low values of 2016 in relation to recent observations. In other words, carbon dioxide decreased further from its maximum values. Likewise, the peak during 2021 is much higher than the peak during 2020. This noise is to be expected though, as it is

Fig. 1 - Carbon Dioxide in Colorado (Niwot Ridge)

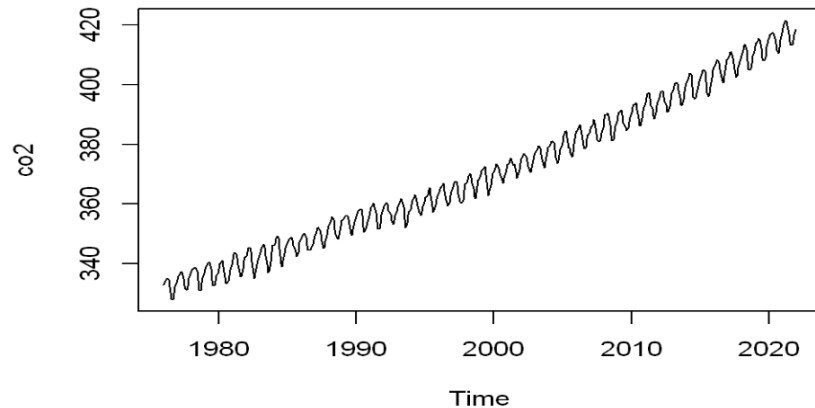
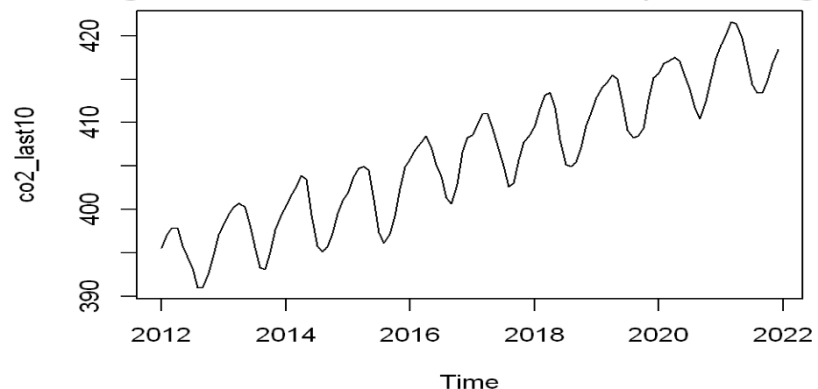


Fig. 2 - Carbon Dioxide in Colorado (Niwot Ridge)



extremely uncommon to see perfect patterns in real-world data. Most of the time, there will be some level of random fluctuation, which relates back to one of the primary motives of time series analysis - to develop a predictive model that minimizes the effect of these fluctuations. The ideal model for carbon dioxide over time would therefore have the lowest possible errors, or difference between actual values and predicted values.

For the sake of simplicity, the analyses conducted in the report will develop models based upon the ten-year dataset plotted in the second figure. This decision was made under the following considerations:

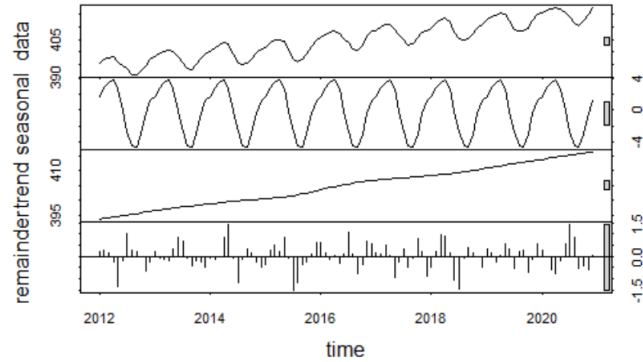
1. The plot for the forty-six-year data shows consistent patterns with respect to trend and to seasonality throughout its duration. Therefore, a subset of the data is expected to show similar patterns. This is supported by the similarity between Figures 1 and 2 in terms of shape.
2. The data seems to fit the fundamental forecasting assumption that patterns will continue in the future, based on the fact that said patterns have persisted decades into the past.
3. While the initial time series may not be independent of time, stationarity is likely achievable through differencing methods. Ideally, the statistical properties of the series should not change over time, including its mean and variance. Differencing methods will be discussed in greater detail in the ARIMA Models section.
4. A ten-year dataset still gives a sample size of 120, which is hypothetically sufficient to uphold the central limit theorem of statistics, in turn allowing for the assumption of normality.

The data set from 2012 through 2021 will be split into a training set and a testing set. This will allow for the computation of forecast errors, which are the primary method of assessing the accuracy of each respective model. The training set will be the first nine years, 108 data points, and the testing set will be the last year, 12 data points. In other words, it will be a 90% to 10% split. Modeling methods will be applied strictly to the training set, and forecasting methods will be applied strictly to the test set.

The following sections will include detailed descriptions of time series regression, exponential smoothing, and ARIMA models. In addition, there will be explanations of the methods and procedures used to develop and select the most appropriate model for each. After this point, the results from each analysis technique will be compared through the calculation of errors. This comparison will allow for the selection of the most appropriate forecasting method overall, the method that is most efficient, accurate, and useful for projecting future values of carbon dioxide.

ii. Time Series Regression

A time series regression model can be written with the predictive variable (y) on the left side and as a function of the explanatory variables. Additionally, the error term to the right side to ensure any random fluctuations are accounted for. To the right is a decomposition of the data and each of the three components modeled separately so we can analyze the individual patterns.



The first step here is to model the trend. The trend reflects the upward or downward slope over time and tells us the general long-term direction of the data (whether it is increasing or decreasing). As we can see from the graph, there is a clear upward trend, meaning there's a trend of carbon dioxide levels increasing with time. This was modeled linearly ($\text{trend} = \alpha_0 + \alpha_1 \cdot \text{time} + \text{error}$) and trigonometrically ($\text{trend} = \alpha_1 \cdot \cos(r \cdot \text{time}/12) + \alpha_2 \cdot \sin(r \cdot \text{time}/12) + \text{error}$). We found that both the models were very similar and the trend line closely matched the data.

Figure 4 - Linear Trend Model

Adjusted R²: 0.9963

Residual S.E.: 0.3982

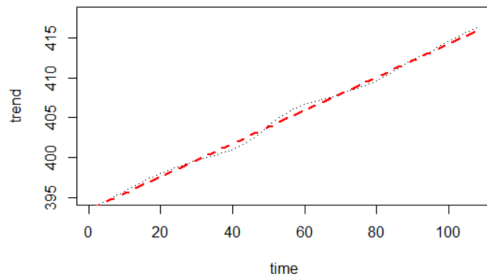
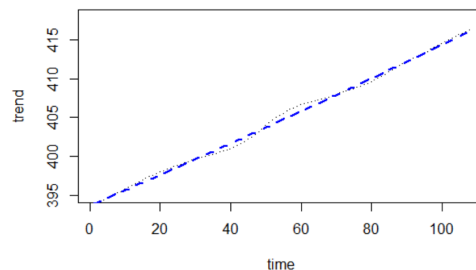


Figure 5 - Trigonometric Trend Model

Adjusted R²: 0.9965

Residual S.E.: 0.3852



Next is seasonality, which shows the repeated up-and-down cycle of the variable as time goes on. This seasonal variation is how the data changes in a specific time frame(week, month, year) and continues throughout the entire data set. In short, it is a repeated sub-pattern found within the data.

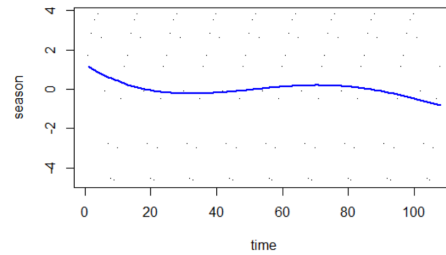
For the seasonality model, we initially tried to fit a polynomial function:

$$fit = \beta_0 + \beta_1 time + \beta_2 time^2 + \beta_3 time^3 + \varepsilon$$

The graph is shown in Figure 6 to the right:

Adjusted R-squared: -0.004907

Residual standard error: 2.975

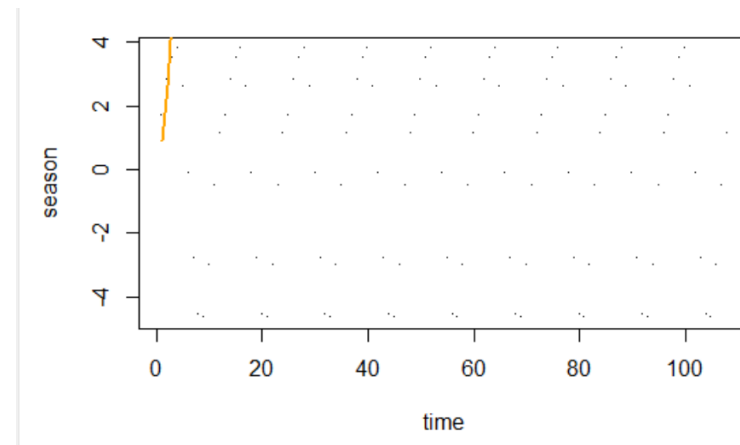


Then with the following trigonometric function:

$$fit = \alpha_1 \cos(a * time) + \alpha_2 \sin(a * time) + \alpha_3 \cos(2a * time) + \alpha_4 \sin(2a * time) + \varepsilon$$

Where $a = 0.017$. This is displayed in Figure 7 below.

Figure 7 - Trigonometric Seasonality Model



Adjusted R-squared: -0.02469

Residual standard error: 3.004

The first polynomial function is clearly a better fit to model seasonality, as we can see from the graphs. It has a higher adjusted r-squared value, meaning a greater percentage of the variance in carbon dioxide levels is explained by the data. There is also a slightly lower residual standard error. Seasonality can also be modeled with dummy variables. Since this

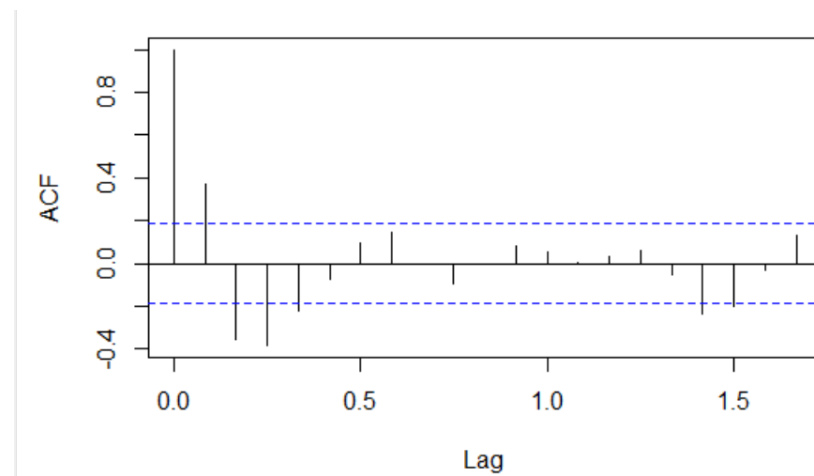
dataset has monthly data, the number of “seasons” or L value is twelve.

After removing the trend and seasonality, all we have left are the random errors in our data. It is not preferred for correlation to exist between the residuals - a good model should have independent residuals with a mean of 0 and a constant variance σ^2 . The covariance between all variables is 0. All autocorrelations are 0 so if the residuals are white noise then we can confirm that there is no correlation between them. We can use the autocorrelation function plot to check if residuals are stationary. This plot has lag k values on the x-axis and autocorrelation on the y-axis. If the data points are between the dashed lines, they are not significantly different from 0. This is known as the Ljung-Box test. Another option is to use the Durbin-Watson method to test for dependence between residuals. This tests the null hypothesis that the correlation is 0, with an alternative hypothesis that the correlation is not equal to 0. A high p-value implies that we have a good model with uncorrelated error terms. For this case, we will move forward with the Ljung-Box test.

Figure 8 shows the autocorrelation function obtained for the residuals in our dataset. There is a correlation because not all lines are between the dashed lines. As we can see, lags 0.1,

0.2, 0.3, 0.4, 1.4, and 1.5 exceed the significance bounds. This tells us that there is significant correlation and non-stationary residuals, which is not an indicator of a good model. Ideally, the residuals should be a white noise process that is independent of time.

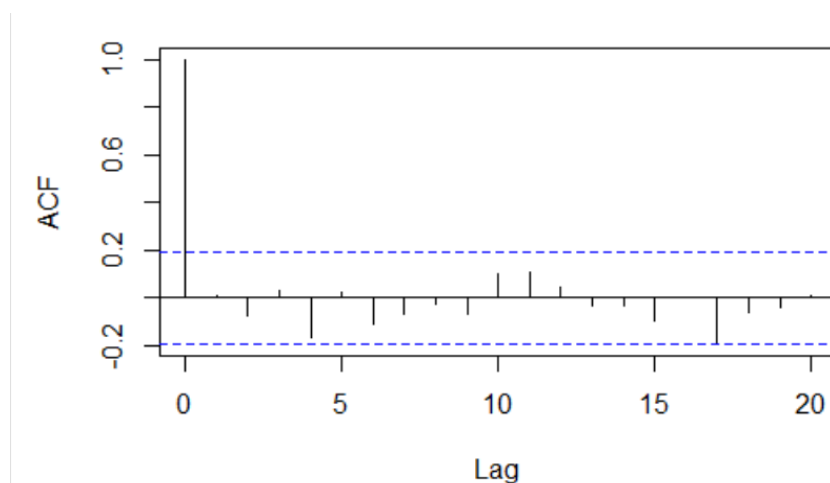
Figure 8 - Initial Autocorrelation Function Plot



To solve this, we fit with AR instead. The residuals were modeled to help with order selection for the autoregressive process. The results suggested that we go up to order 4. To decide whether or not to include the intercept, we compare and see which of the two options have a smaller AIC value. If we include the intercept, the AIC value 112.82 and the second

model without the intercept had an AIC value of 110.84. Therefore, we choose the second model. In the final autocorrelation plot, shown in Figure 9, all lags are in between the dashed lines so we can say that the residuals are stationary.

Figure 9 - Fixed Autocorrelation Function Plot

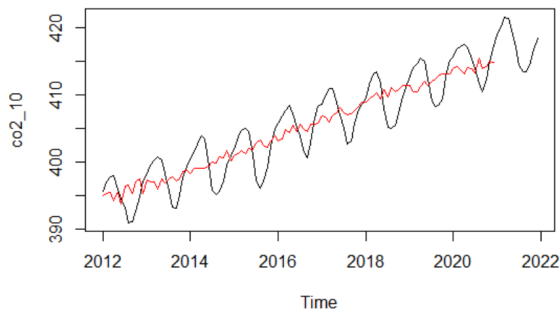


Finally, we modeled all 5 functions created so far with both the trend and seasonality components so we can choose the best one. The model with the lowest mean squared error is considered to be optimal because that means the forecasted values are closer to the actual values. The mean squared error is calculated by the summing squared difference of the

forecasted and actual value and dividing it by the length of data. According to the results below, model 1 is the best model. There were two trigonometric models we created, in addition to the 3 polynomial models below, but they weren't considered because the trend line was extremely inaccurate, such that the graph doesn't display it at all. This indicates that the prediction is too far from the actual values.

Model 1

$$\text{trend} = \alpha_1 \cos(r * \text{time}/12) + \alpha_2 \sin(a * \text{time}/12) + \varepsilon$$

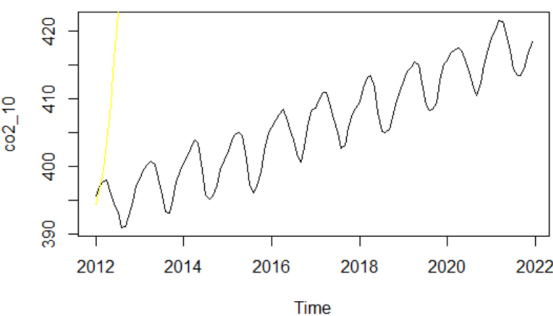


This model seems to capture the linear trend of the data but not the seasonality, resulting in an extremely high mean squared error.

MSE: 1,248,462

Model 2

$$\text{trend} = \alpha_0 + \alpha_1 \text{time} + \varepsilon$$

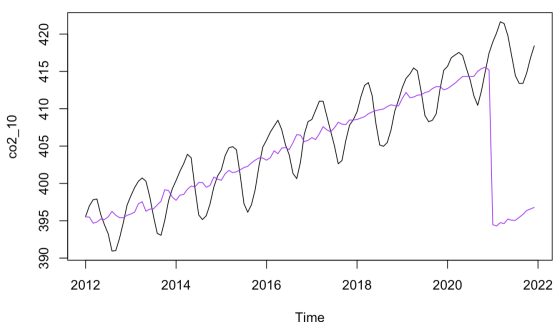


Model 2 seems to entirely miss both the trend and seasonality, leading to an even larger mean squared error. The plot and sheer magnitude of the error instantly rules out this model.

MSE: 130,175,689

Model 3

$$\text{season} = \alpha_1 \cos(a * \text{time}) + \alpha_2 \sin(a * \text{time}) + \alpha_3 \cos(2a * \text{time}) + \alpha_4 \sin(2a * \text{time}) + \varepsilon$$



The third model is extremely similar to the first model until 2021, at which point its predictions decrease dramatically, leading to a significantly larger error metric.

MSE: 2,331,641

Since the first 3 models are not fitting the dataset well, we now try the indicator model which is usually a good try for the seasonality component. The linear and quadratic trends were modeled with dummy variables.

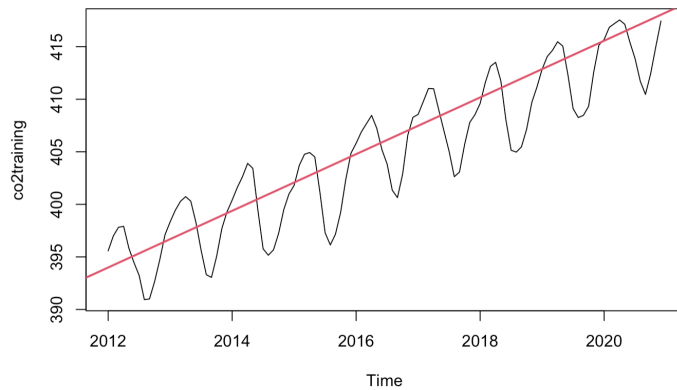


Figure 10 - Linear Trend
Residual standard error: 3.13
Adjusted R-squared: 0.8025

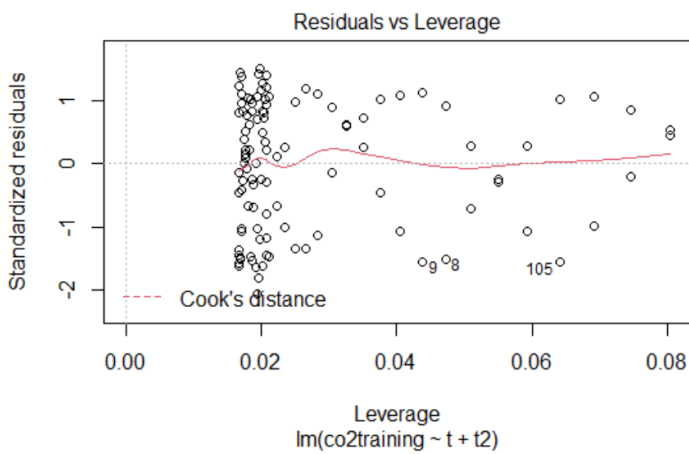


Figure 11 - Quadratic Trend
Residual standard error: 3.142
Adjusted R-squared: 0.8009

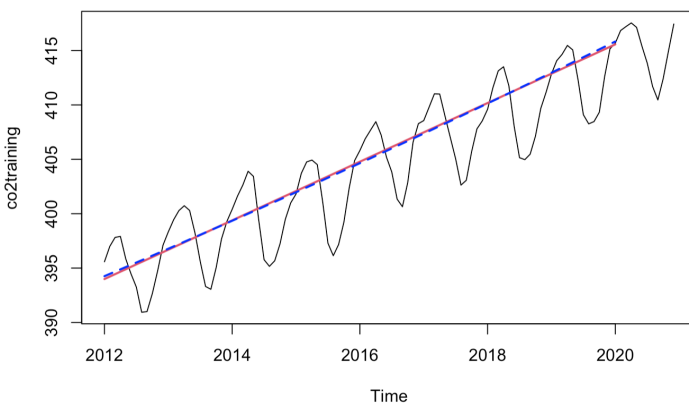
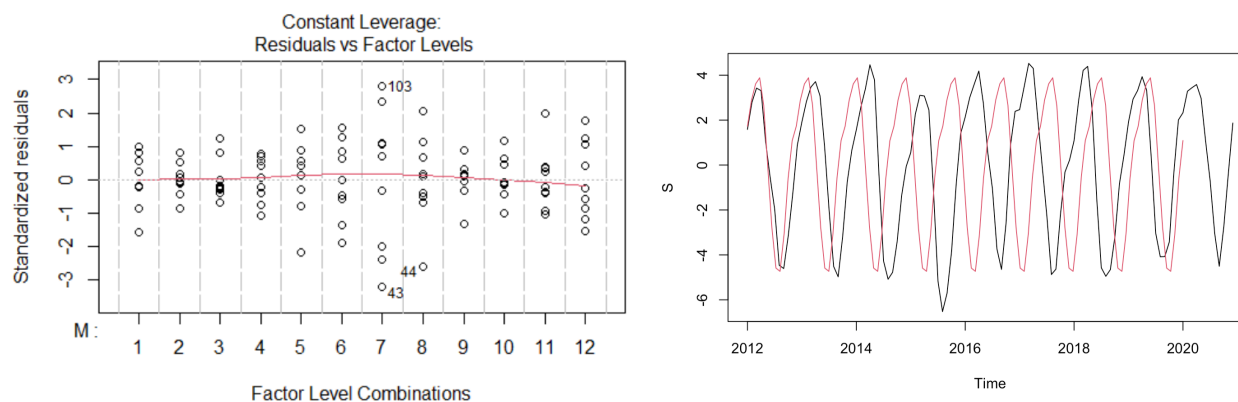


Figure 12 - Quadratic and linear trend combined

According to Figure 12, the Quadratic Trend and Linear Trend have no big difference since the red and blue lines are almost overlapping. We decided to go with the linear trend.

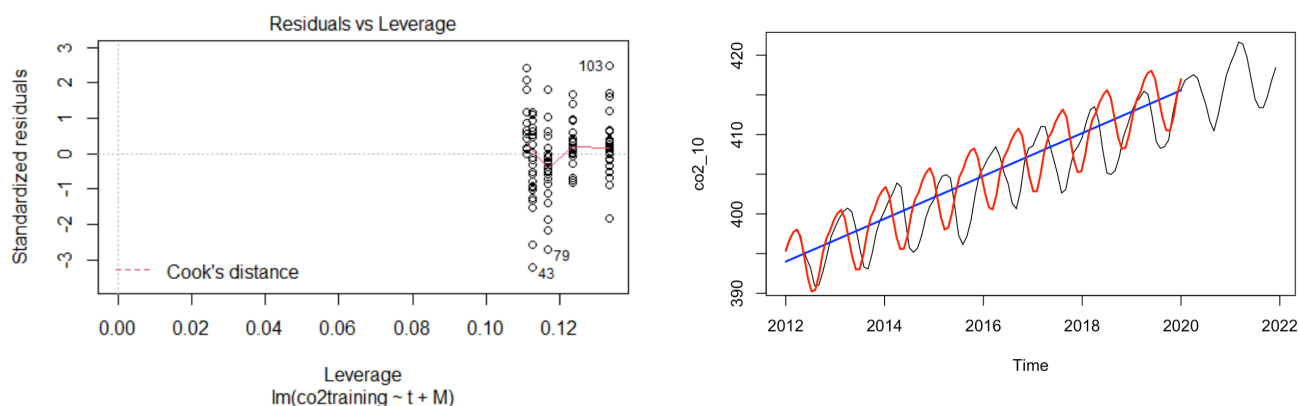
Figures 13 and 14 - Seasonality Model with Dummy Variables



Residual standard error: 0.7601

Adjusted R-squared: 0.9884

Figures 15 and 16 - Trend and Seasonality together



Residual standard error: 0.7491

Adjusted R-squared: 0.8725

As figure 17 shows to the right, the indicator model residual was not stationary, so we modeled the residuals by using AR(5) and attained a σ^2 value of 0.2313. The resulting autocorrelation function can be seen on the following page, vertical lines below the dotted blue lines indicating correlation between

Call:
ar.yw.default(x = res_pre.fit, order.max = NULL)

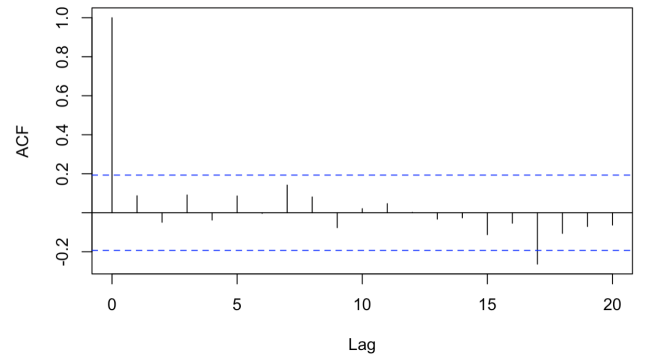
Coefficients:

1	2	3	4	5
1.0242	-0.8585	0.6810	-0.3736	0.2699

Order selected 5 sigma^2 estimated as 0.2313
 AIC Res Model 1: 112.8207
 AIC Res Model 2: 110.8418

certain lag points are not significantly different to zero.

According to the AR summary, 5 orders have been selected, and AIC value for the residual without intercept is 110.8418 (smaller than the one with the intercept). From the ACF, correlogram shows that the sample autocorrelation for the in-sample forecast errors at lag 17 exceeds the significance bounds. As shown in figure 18 below, the p-value of the box test is 0.4393, which is much greater than the critical value 0.05, indicating that there is little evidence of non-zero autocorrelations in the in-sample forecast errors at lags 1-20. Therefore, we can conclude that the indicator model is an adequate predicted model for the CO2 data set.



Box-Ljung test

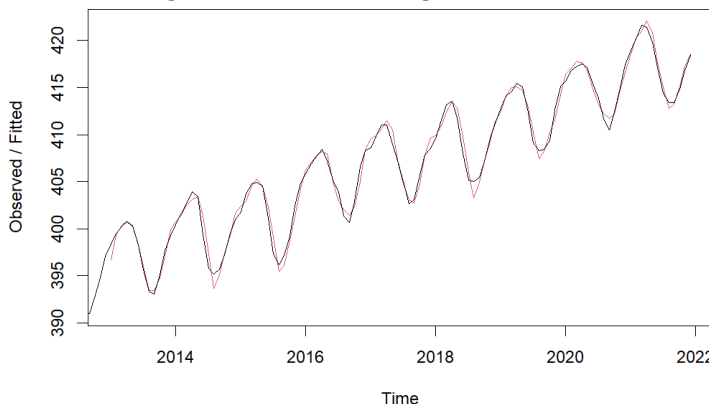
data: resPrefit_ts
X-squared = 20.3, df = 20, p-value = 0.4393

iii. Exponential Smoothing

We used exponential smoothing in the training set of the data (2012-2020) to see the patterns of the carbon dioxide levels clearly and to predict the monthly carbon dioxide levels in the test data set (2021) in order to determine the forecast errors. We also used exponential smoothing for the whole dataset to predict the monthly carbon dioxide levels in 2022.

We plotted the carbon dioxide levels from the whole dataset to determine which exponential smoothing method we should use. We decided to use the Holt-Winters Exponential method because the plots show an increasing trend with some seasonality as well. The Holt-Winters Exponential Smoothing Method estimates three parameters: alpha, beta, and gamma. The alpha parameter is the estimate of the level at the current time point. The beta parameter is the estimate of the slope of the trend component of the time series. The gamma parameter is the estimate for the seasonal component of the time series. All of the parameters are in between 0 and 1. Parameter values that are close to 0 indicate that little weight is on the recent observations when predicting the future values.

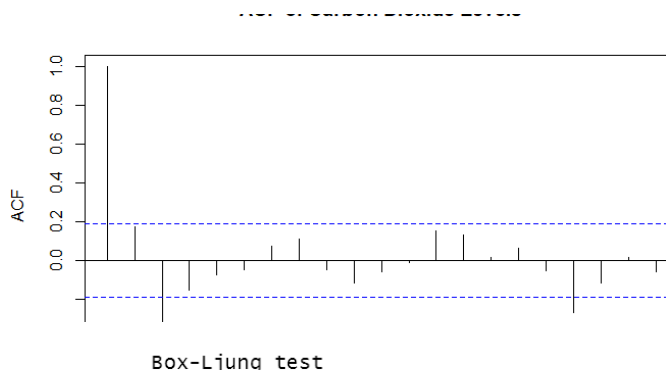
Fig. 1 - Holt-Winters Filtering for the Whole Dataset



From using the Holt-Winters Exponential Smoothing Method for the whole dataset, the alpha level was 0.9073384, the beta level was 0, and the gamma level was 1. Since the alpha and gamma levels are high, this means that the current value of the level and the estimate of the seasonal component of the time point are only based on the recent observations of the carbon

dioxide levels. Since the beta level is 0, it means that the estimated slope of the trend component is not updated through the time series and is set to its original value which is 0. The graph on the right shows the original time series, and the forecasted values from the Holt-Winters filtering. The black line is the original time series, and the red line is the forecasted values from the method. Based on the graph, the method is successful at predicting the seasonality.

However, this method may not be adequate in predicting the carbon dioxide levels in 2022. To determine the adequacy of the method we had to determine if there is correlation in the residuals. If there is non-zero correlation in the residuals, then the model is not adequate in predicting the carbon dioxide levels, and the model needs to be improved. To answer this question, we plotted the residuals and the ACF. We also conducted a Box Test to determine if there is some correlation in the residuals. To the right is the correlogram of the forecast errors for lags 1-20. Based on the ACF plot, the ACF for lags 2 and 17 are outside of the dashed lines,



data: newresid
X-squared = 49.281, df = 20, p-value = 0.0002805

therefore, there are some non-zero correlations in the residuals. We also conducted a Box Test to determine if there are non-zero correlations in the residuals. After conducting the Box Test, we found the p-value to be 0.0002805 which is smaller than the 5% significance level. This means that there is sufficient evidence of non-zero correlations in the residuals. Therefore, this method alone is not sufficient in predicting the carbon dioxide levels and needs to be improved upon by another forecasting technique.

iv. ARIMA Modelling

We used ARIMA models to forecast the carbon dioxide levels in Niwot Ridge, Colorado for 2022 by using the data we got from 2012 to 2021. We tried two different auto ARIMA methods in R, one with seasonal components included and one without. Which gave us two completely different results. The first method was using auto ARIMA to forecast the carbon dioxide levels in 2022 without the season component. We got an ARIMA model of ARIMA(2,1,2) using this method and the acf correlogram of the residuals showed that the model had exceeded the significance bounds several times. The p-value for the box test was also incredibly low (0.0009471). This showed us that the model would do a poor job forecasting and after using the forecast function in R.

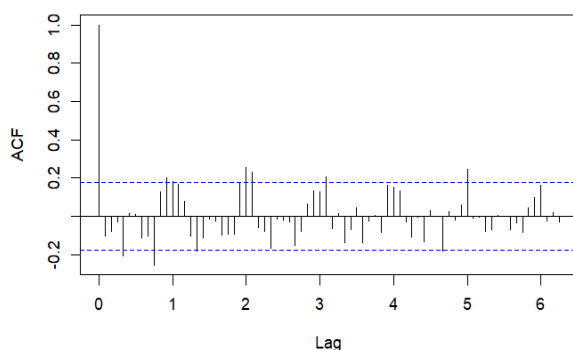


Figure 1

```
Series: co2
ARIMA(2,1,2)

Coefficients:
      ar1      ar2      ma1      ma2
      1.3330 -0.6582 -0.0347 -0.7519
s.e.  0.0715  0.0701  0.0603  0.0573

sigma^2 = 0.7061: log likelihood = -147.83
AIC=305.66  AICc=306.19  BIC=319.56
```

Figure 2

The second and much better ARIMA model was the one with seasonal components included. We got an ARIMA model of $ARIMA(5,0,1)(2,1,0)$. The acf correlogram of the residuals showed that the model had stayed within the significance bounds. And the Box text gave us a very high p-value (0.857). Which meant that we could conclude that there was very little evidence for non-zero autocorrelations in the forecast errors. In a comparison between figure 2 and 4 you'll see that the seasonal ARIMA model includes coefficients for sar1 and sar2. This is because the seasonal ARIMA model uses the autoregressive and moving average values and also takes into account any seasonality patterns. Unlike the first model. Which only takes the past values into account. The second model also has lower AIC values, thus it is much better for forecasting.

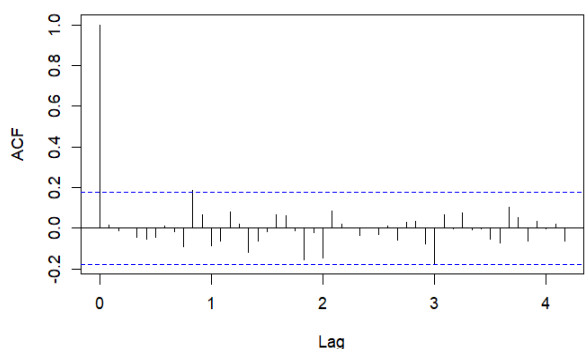


Figure 3

```
Series: co2
ARIMA(5,0,1)(2,1,0)[12]

Coefficients:
      ar1      ar2      ar3      ar4      ar5      ma1      sar1      sar2
      0.7357 -0.2807  0.3023 -0.1190  0.3359  0.7517 -0.6404 -0.2104
s.e.  0.1227  0.1745  0.1631  0.1579  0.1077  0.0983  0.1136  0.1171

sigma^2 = 0.425: log likelihood = -105.73
AIC=229.46  AICc=231.3  BIC=253.6
```

Figure 4

v. Forecasting

Each analysis technique based on the training data was tested using the testing data in order to create the following table of error statistics. Ideally, each method will have very low error statistics, meaning they make accurate predictions. Once this has been confirmed, the models will be used to forecast future carbon dioxide levels.

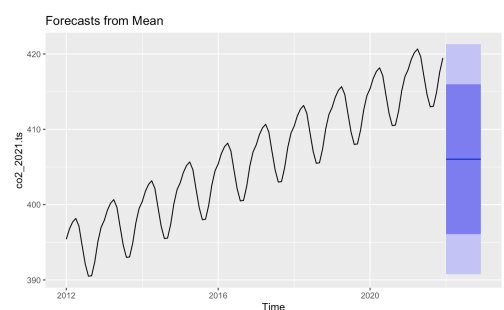
Statistics	Regression	Exp. Smoothing	ARIMA
Mean Absolute Error	6.518	1.059	1.262
Mean Squared Error	2.608	1.471	2.979
Mean Percent Error	-0.036%	-0.19%	-0.004%

The following are results from predicting of 2021 to obtain error statistics and forecasting of 2022 to obtain future values from time series regression. The tables below show the original CO₂ values of 2021 from our dataset and the predicted values for 2021, respectively. These were used to calculate the error statistics provided in the table above.

	CO ₂ <dbl>	t <dbl>	M <dbl>	fit	lwr	upr
109	418.94	2021.000	1	417.8992	416.2839	419.5145
110	420.15	2021.083	2	419.2637	417.6484	420.8790
111	421.65	2021.167	3	420.1837	418.5684	421.7990
112	421.41	2021.250	4	420.6581	419.0428	422.2734
113	419.79	2021.333	5	419.6348	418.0195	421.2501
114	417.10	2021.417	6	417.0670	415.4517	418.6823
115	414.42	2021.500	7	414.5948	412.9795	416.2101
116	413.40	2021.583	8	412.9926	411.3773	414.6079
117	413.40	2021.667	9	413.0515	411.4362	414.6668
118	414.81	2021.750	10	414.9026	413.2873	416.5179
119	416.81	2021.833	11	417.6070	415.9917	419.2223
120	418.42	2021.917	12	419.4770	417.8617	421.0923

Time Series Forecasting

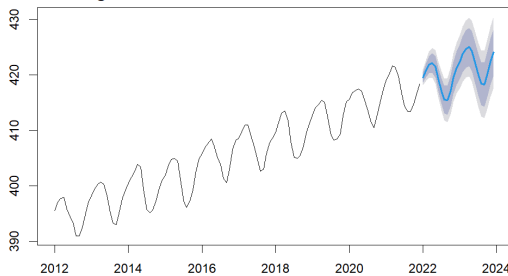
Below is the prediction for 2022 via mean forecasts using the dummy variable model



	Point Forecast <dbl>	Lo 80 <dbl>	Hi 80 <dbl>	Lo 95 <dbl>	Hi 95 <dbl>
Jan 2022	406.0304	396.0893	415.9714	390.7559	421.3048
Feb 2022	406.0304	396.0893	415.9714	390.7559	421.3048
Mar 2022	406.0304	396.0893	415.9714	390.7559	421.3048
Apr 2022	406.0304	396.0893	415.9714	390.7559	421.3048
May 2022	406.0304	396.0893	415.9714	390.7559	421.3048
Jun 2022	406.0304	396.0893	415.9714	390.7559	421.3048
Jul 2022	406.0304	396.0893	415.9714	390.7559	421.3048
Aug 2022	406.0304	396.0893	415.9714	390.7559	421.3048
Sep 2022	406.0304	396.0893	415.9714	390.7559	421.3048
Oct 2022	406.0304	396.0893	415.9714	390.7559	421.3048

While the prediction of a single value for the entire year may not be useful, especially given that the data set shows a clear upward trend along with seasonality, it does in effect give the most conservative prediction possible. There was significant difficulty in forecasting carbon dioxide levels due to the complexity of the residuals - it could therefore be said that time series regression models are not an ideal method of forecasting as it relates to the selected data.

Fig. 2 - Forecasts of Carbon Dioxide Levels in 2022

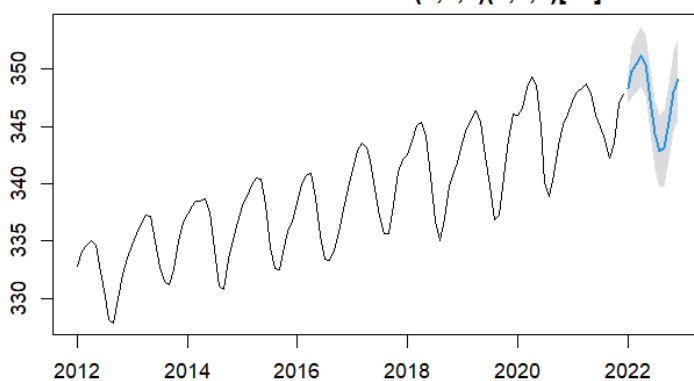


Exponential Smoothing Forecasting:

The graph above shows the forecasted monthly carbon dioxide levels for the year 2022 using exponential smoothing. The actual forecasted carbon dioxide levels are plotted as the blue line. The light blue area of the graph represents the 80% prediction intervals for the carbon dioxide levels. The dark blue area of the graph represents the 95% prediction intervals for the carbon dioxide levels. The predicted carbon dioxide levels seem to follow the same pattern as the previous carbon dioxide levels which is an increasing trend with seasonality.

	Point Forecast <dbi>	Lo 80 <dbi>	Hi 80 <dbi>	Lo 95 <dbi>	Hi 95 <dbi>
Jan 2022	419.5939	418.6500	420.5378	418.1504	421.0374
Feb 2022	420.9276	419.6531	422.2021	418.9784	422.8767
Mar 2022	421.8471	420.3116	423.3826	419.4988	424.1955
Apr 2022	422.2067	420.4486	423.9649	419.5178	424.8956
May 2022	421.4758	419.5201	423.4314	418.4848	424.4667
Jun 2022	419.3071	417.1722	421.4421	416.0420	422.5723
Jul 2022	417.1594	414.8590	419.4597	413.6413	420.6774
Aug 2022	415.6097	413.1552	418.0643	411.8558	419.3637
Sep 2022	415.4476	412.8479	418.0473	411.4717	419.4234
Oct 2022	417.1610	414.4239	419.8982	412.9750	421.3471

Forecasts from ARIMA(5,0,1)(2,1,0)[12]



This graph shows the forecast model of the expected carbon dioxide levels in Niwot Ridge Colorado in 2022 using the ARIMA model with seasonal components included. The blue line is the point forecast at each month and the gray area surrounding it is the forecast range using a 95% prediction interval. The forecast seems to be accurate and captures the increasing then decreasing seasonality pattern that the previous years had.

III - Conclusion

After applying all three forecasting techniques, it is clear that exponential smoothing and ARIMA methods are far more accurate and consistent than time series regression. While time series regression produces a model which is more easily interpreted, it is dependent upon the specific choices made in modeling its three components. This is to say there can be a wide variety of final models when using time series regression, since the data can theoretically be represented in an infinite number of ways. The likely cause of its inaccuracy compared to the other two methods is that the purpose of exponential smoothing and ARIMA is time-series-analysis specific, whereas time series regression is more of an extension of linear regression, leading to more errors. That said, an accurate model for time series regression was obtained through an iterative trial-and-error process. While the method minimized errors, its summary statistics were still larger than those created by exponential smoothing and ARIMA.

Though regression is not as reliable as exponential smoothing or ARIMA, the latter two methods are not without their drawbacks. The limitations of exponential smoothing are clear in that it does not produce an actual model for use in forecasting. Instead, it simply averages select time points within the data in order to “smooth” it out, thus making predictions. While the method led to satisfactory results, there is some level of interpretability that is lost when using this method. Additionally, exponential smoothing provides little to no information about specific variables in an investigation, and can often struggle to make accurate or reasonable predictions far into the future. That said, the method is very useful in making short-term forecasts, has a reasonable amount of flexibility in that its three parameters can be controlled, and that it can be improved through the use of filtering. Another strength of smoothing is that it makes no assumptions about the correlation between successive time points in the data set, meaning it is impossible for any assumption to be violated which would essentially nullify the usefulness of the forecasts produced. Lastly, since there is no explicit model produced by the method, there is no concern for overfitting or underfitting the data. In the case of this report, exponential smoothing is fairly unreliable in that it consistently failed the Ljung-Box test for autocorrelation between residuals. This means that there were significant patterns remaining in the residuals after forecasts were made, which reduces the reliability of predictions because the method is biased toward particular predicted values. That is, the exponential smoothing forecasts may have mispredicted carbon dioxide emissions in a consistent manner, where the ideal method would have normally distributed errors.

The main limitation of ARIMA models is that they are strictly applicable to stationary time series. While stationarity can often be achieved through differencing methods, it is possible for a data set to require a large number of differences, leading to an extremely complex model which in turn reduces interpretability. ARIMA methods could be applied to non-stationary time series, but this would lead to unreliable results if the level and variance of the data appears to change over time. That is, non-stationary models would lead to highly inaccurate forecasts. Despite this, ARIMA can be applied to most time series data sets because it is extremely common for stationarity to be achieved through at most two differences. In addition to this, the simple fact that ARIMA considers the correlation between time points often allows it to produce a better predictive model than time series regression. Its production of an explicit time series model, which allows for non-zero autocorrelation in the irregular component, also increases its interpretability and usefulness in forecasting. This is to say that an explicit forecasting model can hypothetically be used to predict values into the distant future. ARIMA was definitively the most

appropriate method as it relates to the selected data set, as it provided accurate forecasts with extremely low error statistics and passed the Ljung-Box test, removing any concern for autocorrelation between residuals. In summary, ARIMA models gave both accurate and reliable predictions, without being dependent upon a user-defined iterative process.

Generally, it is clear that carbon dioxide values are expected to continue increasing into the future, as all three methods produced forecasts which confirm this. It can also be assumed that the Seasonal pattern will continue as well as the trend. In other words, it is reasonable to predict carbon dioxide levels to reach their lowest levels around the beginning of the fall months, and their highest values around the beginning of the spring months. While this conclusion may be trivial due to the potential causes of increasing carbon dioxide levels discussed in the introduction, the production of several models within the report is certainly of some use. Additionally, the report established a mathematical proof of sorts which suggests that previous patterns are not due to random chance and are therefore likely to continue into the future. Thus, the primary goal of forecasting has been achieved through the methods utilized.

References

- [1] Center for Climate and Energy Solutions. “Global Emissions”
<https://www.c2es.org/content/international-emissions/>.
 [Accessed December 5, 2022]
- [2] Eurostat. “Glossary: Carbon Dioxide Emissions”
https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Glossary:Carbon_dioxide.
 [Accessed December 5, 2022]
- [3] Lan, X., E.J. Dlugokencky, J.W. Mund, A.M. Crotwell, M.J. Crotwell, E. Moglia, M. Madronich, D. Neff and K.W. Thoning (2022), “Atmospheric Carbon Dioxide Dry Air Mole Fractions from the NOAA GML Carbon Cycle Cooperative Global Air Sampling Network”, 1968-2021, Version: 2022-07-28, <https://doi.org/10.15138/wkgj-f215>
https://gml.noaa.gov/dv/data/index.php?category=Greenhouse%2BGases&frequency=Monthly%2BAverages¶meter_name=Carbon%2BDioxide
 [Accessed December 5, 2022]
- [4] National Oceanic and Atmospheric Administration. “Atmospheric Carbon Dioxide Dry Air Mole Fractions”
https://gml.noaa.gov/aftp/data/trace_gases/co2/flask/surface/README_surface_flask_co2.html
 [Accessed December 5, 2022]

Appendix

Alex Caramanico - *Exploratory Data Analysis*

```
colorado_data = read.csv("C:/Users/jacma/Downloads/projectFile.csv")
co2 = ts(colorado_data[,4],start=c(1976,1),frequency=12)
# Figure 1
plot(co2,xlab='Time',main='Fig. 1 - Carbon Dioxide in Colorado (Niwot Ridge)')
co2_last10 = ts(co2[433:552],start=c(2012,1),frequency=12) # last 10 years
# Figure 2
plot(co2_last10,xlab='Time', main='Fig. 2 - Carbon Dioxide in Colorado (Niwot Ridge)')
training = ts(co2_last10[-(109:121)],start=c(2012,1), frequency=12)
test = ts(co2_last10[109:120],start=c(2021,1),frequency=12)
```

Exponential smoothing summary error statistics

```
y = as.numeric(test)
c_forecasts1 = HoltWinters(training, beta=FALSE, gamma=FALSE) # level only
c_forecasts1 # alpha close to 1, only current time point
plot(c_forecasts1,main='Level Only')
c_forecasts2 = HoltWinters(training)
c_forecasts2 # alpha close to 1, beta = 0 gamma = 1,
plot(c_forecasts2,main='Level and Seasonality')
library(forecast)
exp_smo_future1 = forecast(c_forecasts1,h=12) # predict 2021, compare to actual
plot(exp_smo_future1, main='Level Only')
exp_smo_future2 = forecast(c_forecasts2,h=12) # print/plot for forecasts
plot(exp_smo_future2, main='Level and Seasonality')
exp_smo_future2
exp_yhat = exp_smo_future2[[4]] # vector for fitted forecasts
cat('Exp. Smoothing Mean Absolute Error:', sum(abs(y-exp_yhat))/length(y),'\n')
cat('Exp. Smoothing Mean Squared Error:', sum((y-exp_yhat)^2)/length(y),'\n')
cat('Exp. Smoothing Mean Percent Error:', 100*(sum((y-exp_yhat)/y)/n),'\n')
```

Joanne Wang and Jayani Bakshi - *Time Series Regression*

```
alldata = read.csv("C:/Users/jayan/Downloads/projectFile.csv")
head(alldata)
dataco2 <- ts(alldata[,4],start=c(1976,1),frequency = 12)
co2_10 <- ts(dataco2[433:552],start=c(2012,1),freq=12)
co2training <- ts(co2_10[-(109:120)],start=c(2012,1),freq=12)
co2testing <- ts(co2_10[109:120],start=c(2021,1),end=c(2021,12),freq=12)
plot(co2training, ylab="CO2Vol", main = "CO2 in Colorado (2012-2021)", col="black")
decco2 <- stl(co2training,"periodic")
```

```

plot(decco2)
head(decco2$time.series)
tail(decco2$time.series)
season <- decco2$time.series[,1]
trend <- decco2$time.series[,2]
res <- decco2$time.series[,3]
acf(res, main="Series of the Original Residuals")
res_ar <- ar.yw(res, order.max=NULL)
resfit_ts <- ts(res_ar$resid)
acf(resfit_ts[-c(1:4)],main="ACF of residuals from ar_yw fit")
res_fit1 <- arima(res, order=c(4,0,0), include.mean=TRUE)
cat('AIC Res Model 1:', res_fit1$aic, '\n')
res_fit2 <- arima(res, order=c(4,0,0), include.mean=FALSE)
cat('AIC Res Model 2:', res_fit2$aic, '\n')
coef_fit2 <- coef(res_fit2)
res_mod <- arima.sim(list(ar = coef_fit2), sd=sqrt(res_fit2$sigma2), n = 108)
time <- 1:108
dat.trend <- as.data.frame(cbind(trend,time))
r <- 0.018 # Heuristic value
tre_fit1 <- lm(trend~I(cos(r*time/12))+I(sin(r*time/12)),data=dat.trend)
coeff1 <- summary(tre_fit1)$coef
plot(time,trend,pch=".",ylim=c(395,418))
lines(time,coeff1[1]+coeff1[2]*cos(r*time/12)+coeff1[3]*sin(r*time/12),lwd=2,lty=2,col="blue"
)
tre_fit2 <- lm(trend~time,data=dat.trend)
summary(tre_fit2)
coeff2 <- summary(tre_fit2)$coef
plot(time,trend,pch=".",ylim=c(395,418))
lines(time,coeff2[1]+coeff2[2]*time,lwd=2,lty=2,col="red")
fit1=alpha1*cos(a*time)+alpha2*sin(a*time)+alpha3*cos(2a*time)+alpha4*sin(2a*time)+Error
season <- decco2$time.series[,1]
sea_dat <- as.data.frame(cbind(season,time))
a <- 0.017 # 1st to try a=0.017
fit1.ss <-
lm(season~I(cos(a*time))+I(sin(a*time))+I(cos(2*a*time))+I(sin(2*a*time)),data=sea_dat)
coefss1 <- summary(fit1.ss)$coef
plot(time,season,pch='.')
lines(time,coefss1[1]+coefss1[2]*cos(a*time)+coefss1[3]*sin(a*time)+coefss1[4]*cos(2*a*time
)+coefss1[5]*sin(2*a*time),lwd=2,col="blue")
summary(fit1.ss)

```

```

fit2.ss <- lm(season~time+time^2,data=sea_dat)
summary(fit2.ss)
coefss2 <- summary(fit2.ss)$coef
plot(time,season,pch='.')
lines(time,coefss2[1]+coefss2[2]*time+coefss2[3]*time^2,lwd=2,col="orange")
r <- 0.018
tr1 <- coeff1[1]+coeff1[2]*cos(r*time/12)+coeff1[3]*cos(r*time/12)
a <- 0.017
ss1 <-
coefss1[1]+coefss1[2]*cos(a*time)+coefss1[3]*sin(a*time)+coefss1[4]*cos(2*a*time)+coefss1[
5]*sin(2*a*time)
w <- rnorm(length(time),sd=sd(res))
yhat1 <- tr1+ss1+w
d.y1 <- ts(yhat1,start=c(2012,1),frequency=12)
plot(co2_10)
x.axis=seq(2012,2021, length=108)
lines(time,tr1,lwd=2,lty=2,col="green")
r <- 0.018
tr1 <- coeff1[1]+coeff1[2]*cos(r*time/12)+coeff1[3]*cos(r*time/12)
ss2 <- coefss2[1]+coefss2[2]*time+coefss2[3]*time^2
w <- rnorm(length(time),sd=sd(res))
yhat2 <- tr1+ss2+w
d.y2 <- ts(yhat2,start=c(2012,1),frequency=12)
plot(co2_10)
lines(d.y2,col="blue")
tr2 <- coeff2[1]+coeff2[2]*time
a <- 0.017
ss1 <-
coefss1[1]+coefss1[2]*cos(a*time)+coefss1[3]*sin(a*time)+coefss1[4]*cos(2*a*time)+coefss1[
5]*sin(2*a*time)
w <- rnorm(length(time),sd=sd(res))
yhat3 <- tr2+ss1+w
d.y3 <- ts(yhat3,start=c(2012,1),frequency=12)
plot(co2_10)
lines(d.y3,col="red")
tr2 <- coeff2[1]+coeff2[2]*time
ss2 <- coefss2[1]+coefss2[2]*time+coefss2[3]*time^2
w <- rnorm(length(time),sd=sd(res))
yhat4 <- tr2+ss2+w

```

```

d.y4 <- ts(yhat4,start=c(2012,1),frequency=12)
plot(co2_10)
lines(d.y4,col="yellow")
r <- 0.018
tr1 <- coeff1[1]+coeff1[2]*cos(r*time/12)+coeff1[3]*cos(r*time/12)
a <- 0.017
ss1 <-
coefss1[1]+coefss1[2]*cos(a*time)+coefss1[3]*sin(a*time)+coefss1[4]*cos(2*a*time)+coefss1[
5]*sin(2*a*time)
res_mod <- arima.sim(list(ar = coef_fit2), sd=sqrt(res_fit2$sigma2), n = 108)
yhat5 <- tr1+ss1+res_mod
d.y5 <- ts(yhat5,start=c(2012,1),frequency=12)
plot(co2_10)
lines(d.y5,col="purple")
y <- as.numeric(co2training)
sum((y-yhat1)^2/length(y))
sum((y-yhat2)^2/length(y))
sum((y-yhat3)^2/length(y))
sum((y-yhat4)^2/length(y))
sum((y-yhat5)^2/length(y))
t <- seq(2012,2020,length=length(co2training))
t.fit <- lm(co2training~t)
coeff <- summary(t.fit)$coef
summary(t.fit)
plot(co2training)
abline(reg=t.fit,col=2,lwd=2)
t2 <- t^2
t.fit1 <- lm(co2training~t+t2)
coeff1 <- summary(t.fit1)$coef
summary(t.fit1)
plot(co2training)
abline(reg=t.fit1, col='blue', lwd=1)
lines(t, coeff[1]+ coeff[2]*t, lwd=2, col=2)
lines(t,coeff1[1]+coeff1[2]*t+coeff1[3]*t2,lwd=2,lty=2,col='blue')
S <- co2training-t.fit$fit
plot(S)
M <- factor(rep(1:12,length=length(co2training)),levels=1:12)
s.fit <- lm(S~M)
summary(s.fit)
lines(t,s.fit$fit,lwd=1,col=2)

```

```

pre.fit <- lm(co2training~t+M)
summary(pre.fit)
fitD <- lm(co2training~t+M+res_D)
summary(fitD)
plot(co2training)
lines(t,coeff[1]+coeff[2]*t,lwd=2,col='blue')
lines(t,fitD$fit,lwd=2,col='red')
res_pre.fit <- pre.fit$resid
acf(res_pre.fit, main="residual from the Indicator Model")
resPrefit_ar <- ar.yw(res_pre.fit, order.max=NULL)
resPrefit_ar
resPrefit_ts <- ts(resPrefit_ar$resid)
acf(resPrefit_ts[-c(1:5)],main="ACF of residuals from ar_yw fit")
res_fita <- arima(res_pre.fit, order=c(5,0,0), include.mean=TRUE)
cat('AIC Res Model 1:', res_fit1$aic, '\n')
res_fitb <- arima(res_pre.fit, order=c(5,0,0), include.mean=FALSE)
cat('AIC Res Model 2:', res_fit2$aic, '\n')
coef_fitb <- coef(res_fitb)
res_D <- arima.sim(list(ar = coef_fitb), sd=sqrt(res_fitb$sigma2), n = 108)
M1 <- rep(c(1,2,3,4,5,6,7,8,9,10,11,12), length=120)
t1 <- seq(2012, 2021+11/12, by=1/12)
dat<- data.frame(cbind(co2_10, t1, M1))
colnames(dat) <- c("CO2", "t", "M")
data0 <- dat[-(109:120),]
data1 <- dat[(109:120),]
resPrefit_ts <- ts(resPrefit_ar$resid)
acf(resPrefit_ts[-c(1:5)],main="ACF of residuals from ar_yw fit")
Box.test(resPrefit_ts, lag=20, type="Ljung-Box")
fit <- lm(CO2~t+as.factor(M), data=data0)
summary(fit)
co2Predict <- predict(fit,newdata=data1,interval="prediction")
co2Predict.trai <- predict(fit,newdata=data0,interval="prediction")
c.test <- co2Predict[,1]
library(forecast)
data2021 <-
c(395.4046,396.7691,397.6891,398.1635,397.1402,394.5724,392.1002,390.4979,390.5568,392.
4079,395.1124,396.9824,397.9040,399.2685,400.1885,400.6629,399.6396,397.0718,394.5996,3
92.9973,393.0562,394.9073,397.6118,399.4818,400.4034,401.7679,402.6879,403.1623,402.139
0,399.5712,397.0990,395.4968,395.5556,397.4068,400.1112,401.9812,402.9028,404.2673,405.1
873,405.6617,404.6384,402.0706,399.5984,397.9962,398.0550,399.9062,402.6106,404.4806,40

```

```
5.4022,406.7667,407.6867,408.1611,407.1378,404.5700,402.0978,400.4956,400.5544,402.4056,
405.1100,406.9800,407.9016,409.2661,410.1861,410.6605,409.6372,407.0694,404.5972,402.99
50,403.0538,404.9050,407.6094,409.4794,410.4010,411.7655,412.6855,413.1599,412.1366,409.
5688,407.0966,405.4944,405.5533,407.4044,410.1088,411.9788,412.9004,414.2649,415.1849,4
15.6593,414.6360,412.0682,409.5960,407.9938,408.0527,409.9038,412.6082,414.4782,415.399
8,416.7643,417.6843,418.1587,417.1354,414.5676,412.0954,410.4932,410.5521,412.4032,415.1
076,416.9776,417.8992,419.2637,420.1837,420.6581,419.6348,417.0670,414.5948,412.9926,41
3.0515,414.9026,417.6070,419.4770)
```

```
co2_2021.ts <- ts(data2021,start=c(2012,1),freq=12)
```

```
co2_2021.fc <- meanf(co2_2021.ts,h=12)
```

```
autoplot(co2_2021.fc)
```

```
summary(co2_2021.fc)
```

Brooke Springfield - Exponential Smoothing

```
all_data <- ts(gas_data[,4], start=c(2012, 1), end=c(2020, 12), frequency=12)
```

```
carbontsforecasts <- HoltWinters(all_data)
```

```
#Alpha level: 1, Beta Level: 0.001557, Gamma Level: 0
```

```
plot(carbontsforecasts)
```

```
carbontsforecasts2 <- forecast(carbontsforecasts, h=12)
```

```
plot(carbontsforecasts2)
```

```
carbonresid <- ts(carbontsforecasts2$residuals, start=c(2012))
```

```
cresid <- carbonresid[-(1:12)]
```

```
acf(cresid, lag.max=20)
```

```
Box.test(cresid, lag=20, type="Ljung-Box")
```

```
#reject the null. There is significant evidence of non-zero auto correlations at lags 1-20.
```

Tobi Akinyemi - ARIMA Models

```
#install.packages("forecast")
```

```
library(forecast)
```

```
#install.packages("ggfortify")
```

```
library(ggfortify)
```

```
#install.packages("astsa")
```

```
#library(astsa)
```

```
#install.packages("Metrics")
```

```
library(Metrics)
```

```
co2 = read.csv("C:\\Users\\tobij\\Downloads\\projectFile.csv")
```

```
co2 = ts(co2[,4], start=c(2012, 1), end=c(2021, 12), frequency=12)
```

```
head(co2)
```

```
length(co2)
```

```
arimaco2.nonseasonal = auto.arima(co2, allowdrift=FALSE, seasonal = FALSE)
```

```
arimaco2.nonseasonal
```

```

acf(arimaco2.nonseasonal$residuals, lag.max = 75, main = 'Figure 1 - ACF')
Box.test(arimaco2.nonseasonal$residuals, lag = 20, type = "Ljung")

forecastco2.nonseasonal <- forecast(arimaco2.nonseasonal, level = c(95), h = 12)
forecastco2.nonseasonal
plot(forecastco2.nonseasonal)

arimaco2.seasonal = auto.arima(co2, allowdrift=FALSE)
arimaco2.seasonal

acf(arimaco2.seasonal$residuals, lag.max = 50, main = 'Figure 1 - ACF')
Box.test(arimaco2.seasonal$residuals, lag = 20, type = "Ljung")

forecastco2.seasonal <- forecast(arimaco2.seasonal, level = c(95), h = 12)
forecastco2.seasonal
plot(forecastco2.seasonal)

co2_error = read.csv("C:\\Users\\tobij\\Downloads\\projectFile.csv")
co2_error = ts(co2_error[,4], start=c(2012, 1), end=c(2020, 12), frequency=12)

arimaco2.error = auto.arima(co2_error, allowdrift=FALSE)
arimaco2.error

forecast(arimaco2.error, h = 12)
pred_list = c(346.8770,347.8996,348.8563,349.2955,348.3236,
              345.2145,341.7108,339.8953,340.7127,343.1293,345.2366,346.4605)
pred_list

actual_list = c(347.24,348.05,348.33,348.78,347.96,346.04,344.95,343.91,342.27,343.60,
                347.02,347.80)
actual_list

mae(pred_list,actual_list)
mse(pred_list,actual_list)
mape(pred_list,actual_list)

```

Brooke Springfield: Exponential Smoothing

```

gas_data <- read.csv("C:/Users/balex/Downloads/STAT417_files/new_projectFile.csv",
header=TRUE)

```

```
gas_data
tail(gas_data)
head(gas_data)
all_data <- ts(gas_data[,4], start=c(1976, 1), frequency=12)
library("forecast")
ten_yrs <- ts(all_data[433:552], start=c(2012), frequency=12)
plot(ten_yrs, main="Monthly Carbon Dioxide Levels from 2012-2021")
ty_forecast <- HoltWinters(ten_yrs)
plot(ty_forecast, main="Fig. 1 - Holt-Winters Filtering for the Whole Dataset")
#Alpha level: 0.9073384, beta level: 0, gamma level: 1
```

```
tyfcst <- forecast(ty_forecast, h=12)
plot(tyfcst, main="Fig. 2 - Forecasts of Carbon Dioxide Levels in 2022")
tyresid <- ts(tyfcst$residuals, start=c(2012))
plot(tyresid)
newresid <- tyresid[-(1:12)]
plot.ts(newresid)
acf(newresid, lag=20, main="ACF of Carbon Dioxide Levels")
Box.test(newresid, lag=20, type="Ljung-Box")
```